

Capturing the Trends, Applications, Issues, and Potential Strategies of Designing Transparent AI Agents

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ABSTRACT

With the increasing prevalence of Artificial Intelligence (AI) agents, the transparency of agents becomes vital in addressing the interaction issues (e.g., explainability and trust). The existing body of research provides valuable theoretical and practical studies in this field. However, determining the transparency of AI agents requires the systematic consideration of the application categories and automation level, which is hardly considered by the prior literature. We thus apply the bibliometric analysis to gain insights from the published literature. Our work outlines the trend of how the number of studies about AI agent transparency increased over the years. We also identify the major application topics and issues in designing transparent AI agents. Furthermore, we categorize the identified applications according to the specific dimensions (risk and timeliness) and put forward potential strategies for designing different agents. Besides, we suggest the possible transparency degree corresponding to the automation level.

CCS CONCEPTS

• : • **Human-centered computing** → Interaction design; Interaction design process and methods;

KEYWORDS

AI, Transparency, Transparent agent, Trend

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1 INTRODUCTION AND BACKGROUND

Artificial Intelligent (AI) agents, including virtual agents like recommender systems and physical agents like autonomous vehicles [25], are becoming pervasive. The human-like intelligence of AI technology is learned from annotated data rather than pre-set rules. As a result, it is challenging to understand how AI agents work internally and predict AI agents' action, causing HCI issues like distrust and confusion about the performance of the AI agents.

A growing number of researches have noticed the increasingly important role of agent transparency in addressing human-AI interaction issues. In this paper, the term “transparency” refers to the mechanism that can report the behavior and intention of agents[9], the process of decision making, the reasons behind unexpected errors[76], and any other information helping users to understand agents[31]. Determining the transparency of AI agents poses complex challenges, even though there have been numbers of theoretical and empirical studies [5, 31, 75]. While some researchers found that the improvement of the transparency is effective in building trust[63], others argued that increasing the transparency is dispensable [13] or might even lead to information overload [23]. We believe that it is because the desired transparency of AI agents could be various according to the application categories and automation level. For example, the AI clinic diagnosis is a high-risk but non-urgent application, so it requires rigorous context information and an ongoing feedback loop but cares less about information overload and timeliness [73]; the voice assistant, as a low-risk but real-time application, needs to avoid information overload and pay attention to breakdown repair [4]. On the other hand, AI agents such as

autonomous vehicles with different automation levels would also require different transparency levels[7]. Therefore, it is necessary to systematically consider the application categories and automation level when determining the appropriate level of transparency. However, we barely see current meta/review research pay attention to this aspect, let alone summarizing the strategies for designing transparent AI agents with considerations of the application categories and automation level.

Our research aims to capture the trends, applications, issues, and potential strategies of designing transparent AI agents from the existing literature. By adopting the bibliometric analysis, we outline the general trend of how the number of studies about agent transparency increased over the years. Based on the summarized application topics and transparency issues, we propose a two-dimension (risk and timeliness) framework and an automation level framework, helping to identify the potential strategies in operationalizing the transparency of different AI agents.

2 METHODOLOGY

We applied a rationalistic epistemological approach [60] to gain insights from the existing literature. We collected publications about agent transparency from The ACM Digital Library (ACMDL) as the corpus and performed a bibliometric analysis of the corpus to reveal trends, application topics, and issues in agent transparency.

2.1 Data Collection

We searched for the relevant literature in ACMDL using the term “transparency” and its common variations such as “explainability” and “interpretability”. We narrowed down the search results to the field of HCI, and papers must include at least one of the target words (interface, interaction, HCI, UX). The irrelevant literature, like the paper about optical transparency, was manually excluded. After the search and selection process conducted by 2 experts in this field, 1240 relevant papers were chosen. Their metadata including title, keywords, abstract (if available), author names, year, and DOI were extracted for the following analysis phase.

2.2 Bibliometric Analysis

Bibliometric analysis is a quantitative method to carry out literature analysis and has been an integral part of research methodology [32]. It has been used in the field of HCI to reveal research topics and trends, such as the starting places for design innovation with emerging technology [83] and the intellectual landscape of the CHI conference [56].

In our work, to figure out the trend in the number of published papers over the years, we computed descriptive statistics of the metadata and then plotted them to visually illustrate how the number of studies about agent transparency increased in the past (the analyzed result is shown in **Section 3**).

We then applied document clustering [26] to further explore major research topics about agent transparency. The corpus was preprocessed by the following steps: 1) removing stop words that do not convey actual meanings, like “the” and “we”, 2) breaking down the remained words into their roots. For example, the root of “explained” and “explaining” is “explain”. Then we calculated the term frequency-inverse document frequency (TFIDF) [82],

measured the similarity between each paper, and transformed it into a paper-term matrix. According to the matrix, papers were clustered using the K-Means algorithm. Since the number of clusters should be predefined in the K-Means algorithm, we generated three to ten clusters to determine an optimal parameter.

To better understand the topics of clusters, we adopted a semi-automated topic modeling method. Based on the high-frequency words generated by the K-means in each cluster, 4 experts sampled papers that are highly cited or highly relevant to agent transparency for additional reading. During the reading, experts verified and adjusted papers in the clusters, and formed topic words of each cluster. If any paper is deviated from the cluster’s topic or has multiple topics, experts would discuss whether to move it to another cluster. The summarized topic words are shown in Table 1 of **Section 4**.

3 FINDING 1: GENERAL TRENDS ABOUT THE NUMBER OF HCI PUBLICATIONS

Following the general trends about the number of HCI publications shown in Figure 1, we attempt to point out when agent transparency began to receive attention, when the popularity of AI stimulated research interests in agent transparency, and how the interests surged in recent years.

There are a total of 1240 relevant papers in HCI venues including 392 papers related to transparent AI agents. The early HCI publications about the transparency of human-computer interaction can date back to 1983, in which Morland suggested the necessity of explanations in terminal interface design when errors occur [65]. In the same year, Lewis proposed a method to explain the rule-based inference process through an AND/OR tree[54], which could be regarded as the rudiment of agent transparency. Researchers in the following 20 years did not pay much attention to AI agents, but several theories they considered for designing traditional agents, such as psychology [10], could provide a reference for the transparency mechanism of AI agents.

It was not until 2004 that the research focus gradually shifted to the transparency of AI agents. For example, Lee and See identified three dimensions including purpose, process, and performance (3Ps) of AI agents, contributing to the agent transparency and influencing users’ trust calibration [52]. In 2010, as the EPSRC proposed principles of robotics that advise the implementation of transparency in AI agents [70], papers in this field saw an increasing trend. In the same year, the emerging of the ImageNet dataset greatly inspired the following deep learning research. The uncertainty and weak interpretability of deep learning made the transparency of the interactive interface gradually become a concern of HCI researchers. Then there was another significant rise in 2016, owing to the Explainable AI program [36] that emphasizes the explainability of AI agents. Emerging events like Conference on Fairness, Accountability, and Transparency (FAT*) as well as the webinar about transparency and intelligibility led by Microsoft keep this field active.

4 FINDING 2: MAJOR APPLICATION TOPICS AND ISSUES

This section aims to identify the major research topics that prior publications have worked on with respect to transparent AI agents, so as to reveal the common application topics and issues of great

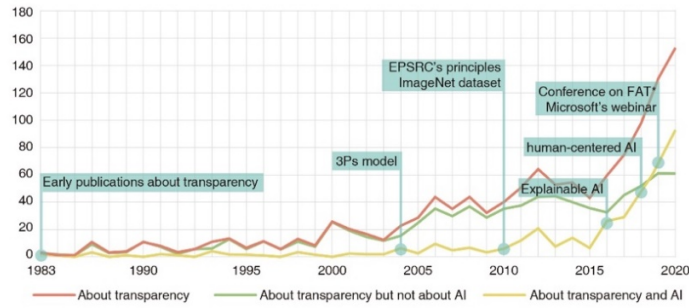


Figure 1: The number of HCI publications mentioned transparency over the years.



Figure 2: The five clusters of the collected publications.

Table 1: The application topics and transparency issues derived from the five clusters

	Application Topics	Transparency Issues
Cluster1	Autonomous driving; Financial decision making;	Trust; Situation awareness;
Cluster2	Clinic diagnosis;	Explanation interface;
Cluster3	Recommender systems; Search engines;	User-centric explanation; Interactive explanation;
Cluster4	Conversational agents; Language generation;	Breakdown repair;
Cluster5	Crime/Education decision making;	Privacy; Fairness;

concern. According to the scatter diagram and high-frequency words generated by K-Means, we conducted the semi-automated topic modeling, and the final topics fell in five clusters as shown in Figure 2. The distance between any two points shows the similarities between the two papers, which is typically known as the semantic distance. We found that the topic words describe each cluster from two perspectives (application topics and transparency issues). Application topics indicate the mainstream applied fields in these clusters, and transparency issues synthesize research focus and challenges mentioned in these clusters. It is noticeable that the clusters are not exclusive because some papers may combine multiple applications and issues. For example, a paper in cluster 3 provides user-centric explanations to support fairness in AI agents [27], and another paper in cluster 1 employs conversational agents in autonomous vehicles to increase user experience [51].

4.1 Application Topics

According to the application topics shown in Table 1, the concern about agent transparency has spread over a variety of fields. Some practice about transparency has been made in designing these

applications. For example, under the autonomous driving topic, Miglani et al. developed two UI concepts for transparency [63] and Haspiel et al. identified the suitable timing of explanation [38]. Efforts are also made in topics of recommender systems and search engines, such as the Aspect-based Transparent Memories method for explaining movie recommendations [28] and Veen Diagram Interface for explaining web search results [49]. When it comes to clinic diagnosis, Sepsis Watch could be a prominent example that could interpret and explain AI models through ongoing feedback loops for hospital clinicians [73]. As for conversational agents and language generation, existing researches identified the lack of transparency in conversational agents [16], and attempted to optimize the user experience through managing the transparency of agents [79]. Apart from that, transparency is much accounted of in decision-making systems in areas such as finance [68], crime [37], and education [5].

4.2 Transparency Issues

Transparency issues in the five clusters can also be referred to Table 1, which contain not only research interests but also unsolved challenges in designing transparent AI agents. The transparency

issues and related research work will be further discussed in **Section 4.2.1 to Section 4.2.5**.

4.2.1 Trust and Situation Awareness. In the first cluster, collected publications mainly discussed transparency issues regarding trust and situation awareness. Trust affects users' perception of whether the agents will perform effectively without a negative or injurious outcome [41]. It would be a key concern in designing human-AI interaction, especially in high-stake situations [45] like autonomous driving. Mignali et al. proposed several factors of transparency to increase the trustworthiness of agents, including continuous feedback of agent status, agent confidence level, predictability of agent's future actions, "why" and "how" information for agent's actions [63]. It is also proved that non-verbal communication [3] and anthropomorphic explanations [51] help to build trust. Additionally, some researchers dug deeper into the relationship between trust and the extent of transparency [43, 45]. Despite the abundant prior literature, there are still challenges such as determining the appropriate level of transparency to avoid over-trust and dis-trust [63, 67]. Kallinen also claimed that transparency influences not only users' trust in agents but also their self-trust [43], which counts a lot in high-risk applications but is rarely considered by current researches.

Situation awareness (SA), another transparency issue in the first cluster, refers to the perception of dynamic environmental elements and events [33]. Based on SA, Chen et al. proposed the Situation-Awareness based agent Transparency (SAT) model to provide guidance on managing information to promote transparency [18]. Furthermore, bidirectional transparency was put forward as an expansion of SAT to promote the human-agent teamwork [20]. Besides, empirical evidence has shown that different extent of SA information could impact not only trust but also users' decision time, workload, and usability [19]. It is noteworthy that the influence of SAT would be different due to the different automation levels, even in similar applications [9, 62, 72, 81].

4.2.2 Explanation Interface. The emerging topic in the second cluster is the explanation interface. Researches in this cluster focus on "What to explain" and "How to explain".

For the question of "What to explain", Amershi et al. provided some generally applicable explanations that are categorized by stages of interaction, including what, why, and how well the agent can do [2]. In addition, Eiband et al. suggested synthesizing the information of experts' mental model and users' mental model [31], which coincides with the advocacy for considering cognitive psychology when designing explanation interface [1]. Lyon's model of transparency stated that the agent should convey its intention, actions, analysis process, and information sensing process to users, and meanwhile present its perception of users [59]. Also, the SAT model provided three levels of transparency, including the agent's perception of the environment, comprehension of the environment and its projection of future status [18]. As for "How to explain", Mittelstadt et al. underlined the importance of post-hoc interpretability, which refers to explaining how the agents behave and why, instead of revealing how a model functions [64]. The applied post-hoc explanation methods include natural language explanations [61], visualization [49], etc. Cai et al. further compared the advantages

and disadvantages of example-based explanations and comparative explanations [17]. Besides, some UI strategies of "white-box" explanation were evaluated in the work of Cheng et al. [21].

4.2.3 User-Centric Explanation and Interactive Explanation. The third cluster reveals the transparency issues about user-centric explanation and interactive explanation, which could supply the design of explanation interface (in cluster 2) as potential solutions. While literature about explanation interface provides general contents and presentation formats of explanation, this cluster focuses more on different user needs.

User-centric explanation pays attention to "why", "when", and "how" different users need transparency information [75], since the need for transparency can be highly context-dependent [75]. As suggested by explainable AI (XAI) and interpretable ML [29, 36], it is essential to provide human-understandable explanations of AI agents. However, compared with the prior works that address transparency from a technical perspective [15, 22, 57], the user-centric explanation, which could bring personal gains and foster the use of agents [35], is preferable by users with low technical expertise when faced with low-risk applications [14] such as recommender systems. Wang et al. proposed a user-centric XAI framework to bridge the gap between algorithm-generated explanations and human cognition [78]. However, though most users prefer brief information [35], it is still essential to provide options for users who want to learn more detailed mechanisms about the agents [14], which brings challenges about more personalized transparency.

On the other hand, the interactive explanation suggests an effective way to implement transparency, which supports users' proactive inquiry about technicalities of agents [86]. Compared with passive feedback from agents, users prefer conversational exchanges with agents [69] and command-based utterances [24, 50], which indicates the advantages of interactive explanation [44]. An example could refer to the work of Zhao et al., which provided a contrastive explanation approach to answering the "Why" and "Why not" questions from users [86]. As a branch of interactive explanation, interactive visualization is widely used in recommendation systems and search engines to help users understand the impact of different algorithms and input data [11, 12, 28]. With the help of the interactive explanation, the implementation of interactive machine learning becomes more efficient with users' increasing awareness of how AI agents make decisions [48]. However, since the interaction is mainly user-initiative, it is likely that agents would fail to respond to the inquiry beyond the scope of their ability, leading to users' unsatisfactory of agents.

4.2.4 Breakdown Repair. Breakdown repair is also a crucial issue according to the fourth cluster. For real-time interactive agents like conversational agents, breakdowns in understanding users' input could frequently happen (especially for the user-initiative interaction). These breakdowns would impact how users perceive and interact with the agents, or even make users give up the task [4]. However, breakdown repair strategies such as confirmation and repeat do not work effectively enough, due to the lack of sufficient transparency of agent status [8], the cause of errors [55], or affordance [58]. To address this challenge, Li et al. designed an interface to help users discover and identify the causes of conversational breakdown, and recover from it [55]. Taking such work as

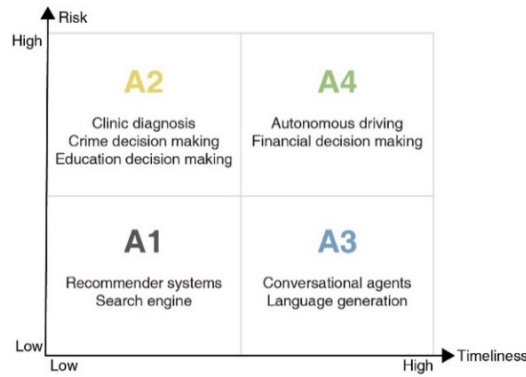


Figure 3: The different types of applications according to the risk-dimension and timeliness-dimension.

inspirations, future work could consider how to integrate adequate transparency into breakdown repair strategies in other applications.

4.2.5 Privacy and Fairness. The fifth cluster reveals ethical issues mainly about privacy and fairness. The privacy issues caused by AI have been increasingly concerned by the academic and industrial communities [74]. For example, in data collection and processing phases, users should have the right to know where the data acquisition device is, what data is captured, how the data is processed, and what the data is used for [80]. In addition, the transparency of AI agents should take both the users and the bystanders (who might be bothered by the interaction) into consideration. Nevertheless, disclosing all the details of AI agents may also infringe on user privacy, making it challenging to seek the balance between privacy and transparency [46]. Future work should also explore how to enable users to personalize the data consent and protect their own privacy [66].

Another issue is about fairness. Fairness requires equity and equality for all groups of users, which is particularly noteworthy in decision-making systems [85]. However, for sociotechnical systems like online rating platforms, the opaque AI algorithms might cause bias, which would damage the fairness or even lead to the loss of trust [34]. Lee et al. claimed that transparency is critical for procedural fairness when designing AI agents, especially in fields like social management [53]. Besides, AI algorithms might result in unfairness when used to infer subjective attributes because many AI algorithms are dependent on subjective data (i.e., crowd-sourced data). For example, the personality judgement based on facial attributes might cause bias in fields like human resources and law enforcement [80]. Therefore, it is still essential to study how to ensure fairness through agent transparency on the basis of principles and measures proposed by prior work [6].

5 APPLICATION CATEGORIES AND POTENTIAL STRATEGIES

The transparency issues are highly context-dependent [75], and we believe that the characteristics of applications could greatly impact the strategies of designing transparent AI agents. Therefore, we extracted dimensions about the characteristics of applications from collected papers. Based on these dimensions, we categorized the

above-mentioned applications through affinity diagramming to find suitable dimensions for designing application-specific transparency. The preliminary dimensions include risk [73, 77], accuracy [39], cost [78], complexity [2, 84], timeliness [39, 47], etc. We critiqued and selected the dimensions according to the following rubrics: the dimensions should 1) be universally applicable for a wide range of applications; 2) effectively differentiate applications in the affinity diagram; 3) be closely associated with the need for transparency. After the selection process, the relevant application topics were finally mapped into four quadrants according to risk-dimension and timeliness-dimension (see Figure 3).

The A1 applications include the recommended systems and etc., whose users would not require extremely timely and accurate feedback. Such applications are less risky because the failure in these tasks hardly leads to serious consequences, and these applications are not expected to adjust learning results in real time. The A2 applications such as the clinic diagnosis would focus more on avoiding risks rather than pursuing timeliness. By contrast, the A3 applications (e.g., the smart voice assistant) are expected to provide immediate responses, and the potential error or misunderstanding would not bring risks. Finally, the A4 applications (e.g., autonomous driving) are required to avoid potential risks when performing real-time tasks.

Faced with the different characteristics of applications, we need to identify corresponding strategies and transparency levels to cope with the related issues. For example, the A2 type agents would pursue the interpretability and trust of the prediction, and help users understand the inference process and results as much as possible to reduce the potential risks; the A3 type agents would pursue the efficiency of predictions, explanations, and breakdown repairs, helping users respond to the external environment as soon as possible; the A4 type agents would explain the process and results of the automatic operation to an appropriate extent while ensuring the efficient information transmission.

6 AUTOMATION LEVEL AND POTENTIAL STRATEGIES

Based on the prior work [9, 62, 72, 81], it is noteworthy that the required transparency levels of agents vary with different automation levels. The AI agents of different automation levels could cooperate

Table 2: The extended SAE Level and the possible transparency level

Level	Description	Possible Transparency Level
0	No automation	The agent only displays the necessary working conditions for assisting fully manual control.
1	Assistance (a single automated agent like cruise control)	The agent displays the continuous feedback of the system and the accessorial information.
2	Partial automation (agent can operate autonomously, but human monitors the environment and can take control at any time)	The agent reports its current/ planned operation, confidence level, and environment information, so that users can take over conveniently.
3	Conditional automation (agents monitor environment, operate autonomously, but human must be available to takeover in some situations)	The agent reports automatic operating status, the oncoming critical situations, predictability of future actions, and corresponding reasons for helping users to supervise the agent.
4	High automation (under certain circumstances, the vehicle is fully autonomous, human takeover is an option in other circumstance)	Only when users are allowed to take over the agent, the necessary information will be displayed to help the user in intervening the agent [42].
5	Full automation (No human interaction required, takeover may be disabled)	Almost no information will be displayed as all information and control are handled by the agent and users would not take over the agent [63].

with humans in different ways. For example, a sweeping robot is supposed to complete the task fully automatically, while the smart voice assistant typically only reports information like weather, traffic jam, etc. according to human's instructions. The Society of Automotive Engineers (SAE) taxonomy of driving automation provides a paradigm that could be extended to other applications. For example, the SAE taxonomy has helped to improve the transparency of conversational technology [30]. Based on the prior work, we extended the descriptions of SAE taxonomy and produced a universal description of SAE Level for the various AI agents as shown in Table 2.

Furthermore, we attempted to suggest the possible transparency level for each level of the SAE taxonomy. To our knowledge, the transparency level will not continuously increase with the automation level of the agent [63]. When the automation level is extremely high, the human might not be able to control the agent, so there is no need for the agent to provide excessive transparency to increase unnecessary workload.

7 THE WAY FORWARD

This paper presents a literature analysis to identify the trends, application topics, issues, and strategies in operationalizing the transparency of different types of AI agents. We are currently illustrating the preliminary analysis of the collected publications. In this paper, we provide a framework to categorize the identified applications according to two specific dimensions (risk and timeliness) and put forward potential strategies for designing different agents. Besides, we propose the possible transparency level corresponding to the automation level.

Future efforts would be made to refine our research considering the potential limitations of our work. As we only collected literature from ACMDL currently, the data from other datasets (e.g., Web of Science) would be added to the corpus for more comprehensive bibliometric analysis. In addition, we would try automatic methods (e.g., X-Means [40]) of finding the optimal parameter and adopt advanced clustering approach (e.g., CSO classifier [71]) to produce

more informative findings. We would also investigate the potential relationship between application categories and the automation level, as well as articulate the corresponding challenges.

Our research can be extended in several aspects. For example, we could hopefully propose the design space of transparency issues, including the format (text, voice, image, etc.), content (why, how, what, etc.), and timing (before tasks, during tasks, after tasks, etc.) of information, through summarizing more actual design cases and academic publications. We could also provide design guidelines based on the two-dimensional application types and extended SAE automation levels, and produce necessary cases or conduct user studies to verify the proposed guidelines. However, the detailed and comprehensive solutions are beyond the scope of this paper and would require research in the future.

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